



Subjective-probability forecasts of existential risk: Initial results from a hybrid persuasion-forecasting tournament^{☆,☆☆}

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ABSTRACT

A multi-stage persuasion-forecasting tournament asked specialists and generalists (“superforecasters”) to explain their probability judgments of short- and long-run existential threats to humanity. Specialists were more pessimistic, especially on long-run threats posed by artificial intelligence (AI). Despite incentives to share their best arguments during four months of discussion, neither side materially moved the other’s views. This would be puzzling if participants were Bayesian agents methodically sifting through elusive clues about distant futures but it is less puzzling if participants were boundedly rational agents searching for confirmatory evidence as the risks of embarrassing accuracy feedback receded. Consistent with the latter mechanism, strong AI-risk proponents made particularly extreme long- but not short-range forecasts and over-estimated the long-range AI-risk forecasts of others. We stress the potential of these methods to inform high-stakes debates, but we acknowledge limits on what even skilled forecasters can achieve in anticipating rare or unprecedented events.

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1. Introduction

Prominent scientists and philosophers have argued that, for a mix of technological and political reasons, the risk of human extinction in this century is higher than at any other point in recorded history (Bostrom, 2019; Ord, 2020; Rees, 2004). This perspective, which we call the “vulnerable century hypothesis”, often highlights the threat of artificial intelligence (AI). Indeed, Ord (2020) sees the AI threat as roughly twice that of all other anthropogenic pathways to human extinction, contributing about 10% to his cumulative estimate of a 16% probability of extinction. Even two of the three “Godfathers of AI”, Geoffrey Hinton and Yoshua Bengio, have expressed serious concern about the existential risks posed by AI (D’Agostino, 2023).

But other observers are less convinced. Pinker (2018) downplays the existential risks from both AI and synthetic biology and has even argued that technology “has not

made this a uniquely dangerous era in the history of our species but a uniquely safe one”, noting our capacity to neutralize natural threats like asteroids. The third “Godfather of AI” and Meta’s chief AI scientist, Yann LeCun, dismisses the threat of misaligned superintelligence as “preposterous” (Kahn, 2023). And prominent AI researchers such as Timnit Gebru and Margaret Mitchell have said the concerns about “imaginary” digital minds are merely a distraction from present AI harms (Gebru, Bender, McMillan-Major, & Mitchell).

This article cannot resolve this controversy, but it does advance a methodology, a multi-stage persuasion-based forecasting tournament (the existential-risk persuasion tournament, henceforth, XPT), designed to improve the rigor of high-stakes debates and track their dynamics. Stage 1 asks forecasting panels of subject-matter experts and talented generalists (“superforecasters”) to make anonymous probability judgments of short- and long-term indicators of existential risks, including AI, nuclear war, bioengineered pandemics, and climate change, with resolution dates for shorter-range forecasts between 2025 and 2030 and those for longer-range forecasts between 2030 to 2100. Stages 2–4 of the XPT tournament ask participants to engage in constructive debates, adversarial collaboration style (Kahneman, 2011), in which each side shares its best arguments and makes good faith efforts to understand the other side’s arguments.

The study we report is preliminary and exploratory: preliminary because we cannot yet assess relative forecasting accuracy for these events and exploratory because we know of no cognitive or social science theory that makes falsifiable forecasts about what to expect when researchers pose such difficult questions to such sophisticated participants. Indeed, it is debatable whether even skilled forecasters are capable of making meaningful probability judgments of outcomes that are decades out and lacking clear historical precedents.

We see this article as the first of a series (based on XPT data) which will grow as evidence accumulates on the accuracy of subjective probability judgments of distant outcomes. Our focus here is on the existential risk debates within the XPT and on the patterns of forecasts across the four stages of data collection as well as across short- and long-range outcomes.

2. Literature review

It is worth emphasizing that the XPT is similar to traditional forecasting tournaments in its research design. For instance, we pose questions that are rigorously resolvable and use proper scoring rules to hold forecasters accountable for their accuracy. These scoring rules treat errors of under- and over-estimation as equally serious (Brier, 1950; Good, 1952; Jose & Winkler, 2009)—a key feature in high-stakes debates in which participants may be tempted to pump up probabilities when they want to mobilize public support for policies (Jervis, 1976; Tetlock, 2005).

But the XPT does more than reward forecasting. It also rewards forecasters for sharing the best evidence and arguments for their judgments, with bigger rewards

accruing to those whose rationales help their peers most (according to their peers). This feature brings forecasters’ incentives into closer alignment with the ideal conditions that Aumann (1976) specified for resolving disagreements among Bayesians. Assuming truth-seeking forecasters with initially diverging views share their most compelling arguments, the pool of common knowledge should expand. Each side will develop a deeper understanding of the other’s reasoning—and as their respective private pools of information shrink, their risk assessments will converge (Aaronson, 2005).

Given the cognitive demands of the exercise and the need to ensure diverging views on AI risk among participants, the XPT selected forecasters from two groups with complementary analytical strengths: (1) AI subject-matter specialists who take the existential risk of AI seriously but often lack extensive forecasting experience and (2) talented generalists or “superforecasters” (Tetlock & Gardner, 2015) who have strong accuracy track records in forecasting but lack AI expertise. These selection criteria were influenced by work on the value of cognitive diversity (Page, 2007). On the one hand, specialists are better positioned than talented generalists to understand technical arguments and the latest developments in their fields; but they are fallible. When experts don’t get regular feedback on their accuracy, they often struggle to translate their causal knowledge into probabilistic predictions and fail to outperform educated generalists or simple algorithms in forecasting tournaments (Kahneman & Klein, 2009; Tetlock, 2017). On the other hand, the talented generalists/superforecasters are a potential check on overconfident experts. To perform well across a wide range of problems, the generalists need an aptitude for determining which experts are good guides to follow. This requires a capacity for open-minded probabilistic thinking, an attentiveness to data, and a tolerance for hearing out clashing perspectives and blending them into predictions. Of course, the generalists are also fallible and have blind spots (Mellers et al., 2015). There is no evidence that their skills extend beyond 12–24 months into the future, a temporal zone where cautious base-rate-grounded forecasting often works well and bold claims about abrupt discontinuities lead to poor accuracy scores.

A natural starting point for data exploration is the range of forecasts that XPT participants endorsed in Stage 1. We organize these data into a $2 \times 2 \times 5$ framework: specialists’ vs. generalists’ forecasts $\times$ catastrophic vs. existential threat forecasts $\times$ forecasts predicated on each of five possible drivers of apocalyptic scenarios (AI, bioengineering, nuclear war, climate change, and non-anthropogenic causation). We did not know, ex-ante, the main or interactive effects of these variables on forecasts, with one key exception. We knew from informal pretesting that the AI subject-matter specialists took AI risks seriously—and that the superforecasters/talented generalists were skeptical of AI risk claims. Beyond that, how each group would view types of existential risk was an open empirical question.

In a similar vein, we had no way of knowing in advance how initial Stage 1 divisions of opinion would evolve across Stages 2–4 of the XPT, a four-month period of platform-facilitated opportunities for each side

to share its best arguments and evidence. However, we were aware that influential schools of thought on human judgment hold diverging views on the potential for convergence in this context. For instance, insofar as the XPT is successful in achieving its goal of facilitating the gradual build-up of a common-knowledge pool among forecasters, proponents of Aumann-style Bayesian convergence should presumably expect gradual shrinkage of disagreements both within and across the specialist and generalist camps.

By contrast, proponents of the heuristics-and-biases school of thought should view convergence as much harder, perhaps impossible to achieve. In this view, we should model XPT forecasters not as truth-seeking Bayesian agents but rather as opinionated agents who apply soft standards to theoretically congenial arguments and tough standards to disagreeable arguments (Gilovich, Griffin, & Kahneman, 2002; Kunda, 1990; Mercier & Sperber, 2011). If this view better captures how XPT participants were thinking, we should expect close-to-zero convergence whenever forecasters can plausibly assert room for reasonable doubt about the diagnosticity of the other side's evidence. It follows that disagreements within and between camps will be remarkably stable across all topics and throughout Stages 2–4 of the XPT.

Previous research on motivated cognition also tells us where to look for forecasters who will be especially resistant to changing their views: those who stake out strong Stage 1 priors on existential risk. Multiple lines of evidence indicate that when people sense more remote risk of being refuted, they make bolder judgments and internalize their positions (Kunda, 1999; Tetlock, 2002). We can therefore expect that individuals with strong initial XPT forecasts will be most prone to confirmatory reasoning, a tendency that is most observable when reality constraints loosen due to severe delays in accuracy feedback, as is true for the longest-range XPT forecasts, which will not be resolved in the lifetimes of most forecasters (e.g., 2100). Here forecasters may opt, consciously or unconsciously, to deprioritize their posthumous reputations and prioritize impressing professional networks with their commitments to in-group claims. Long-range subjective probability judgments now cease to serve the original epistemic function intended by tournament designers and start serving alternative social functions, like affirming in-group loyalty. Far from converging, holders of extreme pre-tournament views will increasingly diverge from moderate forecasters for outcomes increasingly remote in time.

A major problem with evaluating long-range forecasting is that accuracy can only be assessed in the distant future once a forecasting question is resolved. However, the accuracy of intersubjective judgments of long-range forecasters—that is, their forecasts of other forecasters' views—can be assessed immediately. A final research literature, grounded in work on individual differences in human performance, presents the possibility of identifying, ex-ante, the socio-cognitive profiles of the best forecasters over the next few decades via intersubjective forecasting. Research on intersubjective forecasting in general—and Reciprocal Scoring and Bayesian Truth

Serum in particular—implies that: (a) forecasters with extreme pre-tournament views will be less adept at predicting other participants' forecasts (Karger, Monrad, Mellers, & Tetlock, 2021; Prelec, 2004; Prelec, Seung, & McCoy, 2017); (b) forecasters who know more about what others think about the future will offer more accurate assessments of the future. To explore these ideas, all XPT participants were asked on 14 questions not only to predict the objective outcome, what will actually occur, but also to predict the forecasts of other forecasters, both the group to which they belong (fellow specialists or generalists) and the group to which they do not.

The intersubjective forecasting metrics will also shed light on the sincerity of extremists' forecasts of distant futures. Consider two possibilities: (a) if forecasters truly believe their extreme long-range forecasts on objective outcomes, they will have good reasons for continuing to make equally extreme long-range forecasts on intersubjective outcomes that can be immediately scored for accuracy. This argument suggests that these forecasters do not realize how extreme their objective forecasts are and have fallen prey to the naive-realism effect: they assume they see the world as it is and therefore assume wrongly that others will see it the same way (also known as the false-consensus effect) (Gilovich, 1990; Ross, Greene, & House, 1977); (b) if forecasters do not truly believe their extreme long-range forecasts on objective outcomes, they will have good reasons to moderate their long-range forecasts of what others will think (intersubjective metrics), since embarrassing errors on an intersubjective dimension can be detected almost immediately. This argument suggests that extreme forecasters do realize how extreme their objective forecasts are and can locate themselves accurately in the opinion distribution.

In sum, data from the XPT, as of 2024, allows us to explore both content and process questions: content questions about the patterns of existential risk forecasts made by specialists and generalists and process questions about how these patterns of judgments change as a function of progress through the four stages of the XPT, as well as across shorter- and longer-term forecasting horizons.

3. Methods

Of the 198 participants who signed up for the XPT—a four-month exercise from June through October 2022—169 submitted forecasts. The main research sample consists of 89 superforecasters (high scorers in previous tournaments) and 80 subject-matter experts in different areas of existential risk. A total of 111 completed all stages of the tournament.¹

To recruit experts, we contacted organizations working on existential risk, relevant academic departments, and research labs at major universities and within companies operating in these spaces. We also advertised via blogs

¹ Of the 111 forecasters who completed all four stages of the tournament, 72 were superforecasters and 39 were experts. Although 111 completed all stages of the tournament, we report data from forecasters who attrited from the tournament in relevant analyses below.

and Twitter. We received more than 500 expressions of interest, screened these respondents for expertise, and offered slots to the best qualified after a review of their backgrounds.² Our final expert sample (N=80) included 32 AI experts, 15 “general” experts studying long-run risks to humanity, 12 biorisk experts, 12 nuclear experts, and 9 climate experts, categorized by the same independent analysts who selected participants. Our expert sample included well-published AI researchers from top-ranked industrial and academic labs, graduate students with backgrounds in synthetic biology, and generalist existential risk researchers working at think tanks, among others. According to a self-report survey, 44% of experts spent more than 200 h working directly on causes related to existential risk in the previous year, compared to 11% of superforecasters.³ The sample drew heavily from the Effective Altruism (EA) community:⁴ about 35% of experts and 7% of superforecasters reported that they had attended an EA meetup.⁵ This is unsurprising given the EA community’s interest in existential risk (and willingness to participate in projects forecasting those risks).

To recruit superforecasters, we worked with Good Judgment Inc. (GJI),⁶ a company built on earlier academic work by Mellers et al. (2014) that provides paid forecasting services to clients. Most of the recruited superforecasters were directly labeled as such via earlier academic work. A minority of these superforecasters were not labeled as such via that work but were instead top performers in subsequent short-run forecasting tournaments run by GJI. While superforecasters are a set of forecasters with high levels of accuracy on short-run (0–2 year timespan) resolvable questions, it is an open question whether forecasters who are accurate on short-run questions will also be accurate on longer-run questions. We cannot answer that question here but will revisit it when our questions begin to resolve in 2024 and 2030.

Participants moved through a four-stage deliberative process that monitored the evolution of their forecasts and explanations.⁷ This multi-stage process incentivized forecasters not only to make accurate predictions but also to provide persuasive rationales to help peer participants produce well-explained forecasts. As detailed

below, the multi-stage design of this process was heavily inspired by Delphi processes (Rowe & Wright, 2001) but deviated from traditional Delphi methods in significant ways. In particular, forecasters were not necessarily anonymous beyond Stage 1, and forecasters were given access to each other’s forecasts and rationales at earlier stages than in many Delphi processes. The reason for many of these deviations has to do with expert forecaster retention; several experts who expressed interest in the tournament suggested that they would be most interested in participating if a feature of the tournament was frequent collaboration and discussion with other experts. We revisit these design issues in the Discussion Section.

During Stage 1, participants worked individually to submit forecasts and rationales on 14 required and 45 optional short-run and long-run questions concerning global risks. The risks covered AI, pandemics (both natural and engineered), nuclear weapons, and non-anthropogenic risks like volcanoes or asteroids. For each risk area, we asked about both the probability of human extinction and of a catastrophic event that kills at least 10% of the population.⁸ Illustrative short-run questions include:

- What will be the global surface temperature change as compared to 1850–1900, in degrees Celsius? (By 2030, 2050, 2100)
- By what year will fusion reactors deliver 1% of all utility-scale power consumed in the United States?
- How much will be spent on compute [computational resources] in the largest AI experiment? (By 2024, 2030, 2050)
- What is the probability that artificial intelligence will be the cause of death, within a 5-year period, for more than 10% of humans alive at the beginning of that period? (By 2030, 2050, 2100)
- What is the overall probability of human extinction or a reduction in the global population below 5,000? (By 2030, 2050, 2100)

For the longer-run questions resolving after 2030, we asked participants to not only submit a forecast but also to predict the median forecast on that question among both experts and superforecasters (separately).⁹ The long-run questions were required, and nearly all participants made a forecast for each of them. The questions resolving sooner were not all required. Instead, participants forecasted on around five randomly selected short-run questions with the option of forecasting on any number of additional questions. As a result, fewer participants submitted forecasts on each short-run question. When we report probabilities of long-run catastrophic and existential

² Two independent analysts categorized applicants based on publication records and work history. When the analysts disagreed, a third independent rater resolved disagreement after a group discussion.

³ 1% of experts and 2% of superforecasters did not respond to this question.

⁴ Major organizations associated with Effective Altruism describe the movement as a research field and practical community that “aims to find the best ways to help others, and put [those ways] into practice” (Center for Effective Altruism, “Introduction to Effective Altruism”). There are multiple, often-overlapping subcommunities within EA, including groups focused on “longtermism” and “AI safety”, both of which tend to focus on extinction risks and are thus particularly relevant for this study.

⁵ 16.25% of experts and 8% of superforecasters did not provide an answer to this question.

⁶ See <https://goodjudgment.com/>.

⁷ Participants consented to participate in this study, via the University of Pennsylvania’s Institutional Review Board. The consent form details the format of the study.

⁸ We define a catastrophic event as one causing the death of at least 10% of humans alive at the beginning of a five-year period. We define extinction as reduction of the global population to less than 5,000.

⁹ We asked the core required questions about existential and catastrophic risks over 2030, 2050, and 2100 time horizons. Over each of these time horizons, including 2030, we asked participants to submit their own forecast and their forecasts of others’ beliefs. On all other questions resolving by 2030, forecasters submitted only one (properly incentivized) forecast.

The screenshot shows the XPT forecasting platform interface. On the left is a navigation menu with a 'User Stage' dropdown set to '1'. The menu items include: ABOUT, PROGRESS TRACKER, ADMIN, Filter questions by tags, Payment details, 1. Genetically Engineered Pathogen Risk, 2. Non-Genetically Engineered Pathogen Risk (highlighted), 3. AI Catastrophic Risk, 4. AI Extinction Risk, 5. Nuclear Catastrophic Risk, 6. Nuclear Extinction Risk, 7. Non-Anthropogenic Catastrophic Risk, 8. Non-Anthropogenic Extinction Risk, 9. Total Catastrophic Risk, 10. Total Extinction Risk, 11. Year of Extinction, and a 'View More' link.

The main content area displays a question: "What is the probability that a non-genetically-engineered pathogen will be the cause of death, within a 5-year period, for more than 1% of humans alive at the beginning of that period...". Below the question are three radio buttons for time periods: "...by the end of 2030?", "...by the end of 2050?", and "...by the end of 2100?".

Under "Question and resolution details", it defines a pathogen and a naturally arising pathogen, and states that the 5-year period must end before the resolution date. It also mentions that if reasonable people disagree, the question will be resolved via a panel of experts.

A link "For more details and relevant resources, click this link" is provided. Below is a "Clarifications" section with two bullet points: "6/20/22 The relevant 5-year period must end before the resolution date, and begin on or after June 15th, 2022" and "6/30/22 For this question, a 'naturally-arising pathogen' does not include any pathogen created via genetic modulation or manipulation by any intelligent agent, including AI."

At the bottom, there are input fields for "Add tags" and buttons for "2030", "2050", "2100", and "Biosecurity", each with a "Remove" link. Below these is an "EDIT" section with the text: "Please enter your forecasts as percentages below. Please answer for your own beliefs, your beliefs of what other experts will say, and your beliefs on what other superforecasters will say." There are three tabs: "Your Beliefs", "Other Experts' Beliefs", and "Other Superforecasters' Beliefs".

Fig. 1. An image of the XPT forecasting platform, showing the question, “What is the probability that a non-genetically-engineered pathogen will be the cause of death within a 5-year period for more than 1% of humans alive at the beginning of that period?” resolution details, and space to forecast the question for the periods ending in 2030, 2050, and 2100

risk in this report, we report forecasters' own (unincentivized) beliefs, but we rely on the incentivized forecasts to calculate measures of intersubjective accuracy.

We informed all forecasters of the scoring ground rules: questions resolving by 2030 were scored using traditional metrics where the goal was to minimize the gap between probability judgments and reality (coded as zero or one as a function of the outcome). However, for the long-run questions, participants learned that they would be scored based on the accuracy of their reciprocal forecasts: the better they predicted what experts and superforecasters would predict for each question, the better their score. (For more detail, see Karger et al., 2023.) Almost all participants answered the key long-run questions about existential and catastrophic risk. For questions about shorter-run predictors of those risks and the mechanisms underlying them, we asked a smaller, random subset of participants to respond, while all other participants had the option to answer. Fig. 1 shows the platform used by participants to complete these tasks.

One common challenge in forecasting tournaments is to uncover the reasoning behind predictions. To that end, the XPT offered ten \$1,000 prizes for the most persuasive

individual forecast rationales that enhanced others' forecasting accuracy. We explained that after the tournament, we would show the highest-quality anonymized rationales (curated by independent readers) to panels of online survey participants who would make forecasts before and after reading the rationale. Prizes go to those whose rationales helped participants update their forecasts toward greater accuracy, using both proper scoring rules for resolvable questions and intersubjective accuracy for unresolvable questions.¹⁰

In Stage 2, participants worked in teams of 16 who could see each other's forecasts and rationales and update them based on what they learned from teammates.¹¹ Each team consisted entirely of superforecasters or experts. Teams of experts were block-randomized to contain a roughly equal proportion of experts from each

¹⁰ At this juncture, we are in the process of calculating these prizes, but we have already allocated separate prizes to participants for high-quality comments and high-value teammates.

¹¹ Forecasters used usernames and could choose to be anonymous. Most used their real name or otherwise referred to their identity in their forecasts, although that choice was not universal.

domain of the tournament (AI, biorisk, nuclear war, and climate). Forecasters were encouraged to discuss differences of opinion¹² and update their forecasts if their beliefs changed. We offered an additional ten \$1,000 prizes for the highest-quality comments forecasters offered in response to teammates’ rationales and comments from Stage 1.¹³

In Stage 3, superforecasters and experts were blended into integrated teams of 32 and encouraged to deliberate and update their individual forecasts using the same scoring rules as in Stage 1.¹⁴ Each integrated team was asked to combine their rationales into a team wiki¹⁵ describing the strongest argument for the team’s median forecast, along with rationales for lower- and higher-end forecasts that they considered plausible. They were also asked to describe key areas of disagreement and uncertainty. (Each team received \$100 for each question, awarded according to how much each member contributed to the wiki.)

In Stage 4, each team was shown a randomly selected wiki from another team and asked once more to reconsider their forecasts.

In summary, the stages and timeline of the XPT were:

Stage description	Stage duration
(1) Participants made individual forecasts	June–July (~4 weeks)
(2) Teams comprised entirely of either superforecasters or experts deliberated and updated their forecasts	July–August (~6 weeks)
(3) Blended teams from the second stage, consisting of superforecasters and experts, deliberated and updated their forecasts	August–September (~5 weeks)
(4) Each team saw one wiki summarizing the thinking of another team and again updated their forecasts	October (~4 weeks)

After the conclusion of the XPT, participants filled out a postmortem survey on their experience in the tournament, their views on global risks, and where they would allocate funding to address those risks.

Finally, we surveyed a wider segment of the public about their views on existential risks. The survey included 912 college graduates, largely from the United Kingdom and United States, who had engaged in previous forecasting-related surveys and were selected using the survey platform Prolific. While the respondents are not a representative sample, the survey helps place the views

of the superforecasters and existential risk experts in the context of wider public opinion. We gave these public respondents a subset of the 59 questions from the tournament, including all long-run questions. Approximately 500 members of the public responded to each of the core questions.

4. Results

Table 1 provides the median forecast of XPT participants on whether AI, use of bioweapons, nuclear war, or non-anthropogenic risks will, by 2100, cause a "catastrophe," killing within a five-year period more than 10% of humans alive at the beginning of that period. We separately asked experts about five different types of risk. Relative to generalists, specialists expressed greater concern about both catastrophic and existential risk across the board, with the differences most pronounced on AI-related risk.

Table 2 presents forecasts of extinction risks, indicating whether, by 2100, forecasters believe each risk will cause humanity to go extinct. We see similar patterns, with experts more concerned about risks in their subfield relative to non-experts and specialists. While nuclear weapons are seen as the most likely cause of a catastrophic event among all forecasters, AI is seen as the most likely source of human extinction.

Superforecasters put a considerably lower probability (than experts) on human extinction by 2100: 1% vs. 6%, respectively. They also put lower chances of catastrophe and extinction in each of the risk areas studied in the XPT, with two exceptions: non-anthropogenic risks, where the two groups provide virtually identical forecasts; and catastrophic risk from natural pathogens. See Appendix and Figs. A.1 through A.3 for boxplot figures highlighting this large variation in beliefs among subgroups.

5. Exploring ideas about judgment processes

From an Aumannian/Bayesian perspective, disagreements should shrink when reasonable people with initially opposing views have more opportunities to understand each other’s reasoning. However, we do not find this pattern when plotting each group’s median forecasts for total extinction risk (by 2100) across different stages of the tournament. Fig. 2 shows that although there was some within-school-of-thought convergence among AI experts in Stage 4 of the XPT, there was no sign of superforecasters’ and experts’ views of total extinction risk moving toward one another across stages of the tournament. Similar patterns emerged on the other categories of catastrophic and extinction-risk questions.

The motivated cognition perspective, however, expected stability of disagreements within and between camps, which is not obvious in Fig. 2. This stability in opinions becomes more visually apparent when we examine the changing variance in forecasts within each group across each stage, relative to the initial level of variance (normalized to 100% on day 1 of the tournament). Fig. 3 shows that although there was modest

¹² Most discussion occurred asynchronously on the platform, but some teams coordinated phone or Zoom calls.

¹³ Prize-winning comments were selected based on votes from the participants, who were asked to select the three most helpful comments that their teammates had posted throughout the tournament.

¹⁴ Each team of 32 forecasters consisted of one team of 16 superforecasters from Stage 2 and one team of 16 experts from Stage 2.

¹⁵ "Wikis" refer to collaborative documents created by each team during the study. A team’s wiki served as a platform for team members to crowdsource the collective rationale behind their forecast.

Table 1

We calculate medians from N=88 superforecasters; N = 66 domain experts with expertise in AI, nuclear, and biorisk domains; N = 14 general x-risk experts; N = 482 members of the general public. Causes are not necessarily mutually exclusive so our resolution criteria allow that if, for example, an AI system uses nuclear weapons to cause a catastrophe in a manner that requires both technologies, it would count as both AI- and nuclear-caused catastrophe. We also present bootstrapped confidence intervals for each median.

Forecasts of Catastrophic Risk by 2100					
Forecast	Median Estimate (95% confidence interval)				
	Superfore-casters	Domain experts	Non-domain experts	General x-risk experts	Public survey
AI catastrophe	2.13% [1.80, 3.00]%	12% [4, 17.5]%	6.16% [5, 12]%	10% [6.10, 16.12]%	5% [5, 5] ^a
Engineered pathogen catastrophe	0.85% [0.5, 1.0]%	3% [1, 5]%	3% [1.5, 5.0]%	5% [3.03, 10.00]%	–
Natural pathogen catastrophe	1% [0.7, 1.3]%	0.8% [0.5, 1.0]%	1.5% [1, 3]%	1.9% [0.2, 2.5]%	–
Nuclear catastrophe	4% [3, 5]%	8% [5, 11]%	8% [7, 10]%	7.24% [4.9, 10]%	10% [10, 11]%
Non-anthropogenic catastrophe	0.05% [0.033, 0.081]%	0.09% [0.05, 0.11]%		0.045% [0.01, 0.97]%	2% [1, 3]%
Total catastrophic risk	9.05% [6.12, 10.2]%	18.5% [13.9, 25.0]%		28.95% [18.70, 50.63]%	11.56% [10, 13]%

^a This confidence interval may appear incorrect but so many people in the public submitted forecasts of 5% that the median is stably 5% across bootstrapped draws from the sample.

Table 2

We calculate medians from N = 88 superforecasters; N = 66 domain experts with expertise in AI, nuclear, and biorisk domains; and N = 14 general x-risk experts.

Forecasts of Extinction Risk by 2100					
Forecast	Median Estimate (95% confidence interval)				
	Superfore-casters	Domain experts	Non-domain experts	General x-risk experts	Public survey
AI extinction	0.38% [0.10, 0.75]%	3% [0.49, 10.00]%	2% [1.00, 4.03%]	4.75% [2, 14.0]%	2% [1, 2]%
Engineered pathogen extinction	0.01% [0.003, 0.054]%	1% [0.1, 1]%	0.1% [0.05, 0.30]%	1% [0.12, 1.09]%	–
Natural pathogen extinction	0.0018% [0.001, 0.005]%	0.01% [0.0005, 0.0200]%	0.008% [0.0001, 0.0640]%	0.001% [0.0001, 0.2000]%	–
Nuclear extinction	0.074% [0.022, 0.100]%	0.55% [0.075, 1.400]%	0.19% [0.08, 0.50]%	0.7% [0.014, 1.000]%	2% [1.5, 4]%
Non-anthropogenic extinction	0.0043% [0.0025, 0.007]%	0.004% [0.0015, 0.0070]%		0.0059% [0.0010, 0.0095]%	1% [0.5, 1.0]%
Total extinction risk	1% [0.65, 1.23]%	6% [3.00, 10.75]%		6.6% [3.001, 13.670]%	5% [3, 5]%

intra-group convergence, as measured by a reduction in variance after Stage 1 (independent forecasting), forecasters largely maintained their positions after Stage 2. The standard deviation of generalist/superforecaster judgments was maintained at nearly 90% of its value at the end of Stage 1—and the standard deviation of expert forecasts was maintained at 95% of its initial value. In brief, forecasters' opinions on existential risk topics remained relatively stable throughout the tournament.

To measure pre-tournament views, we partitioned forecasters into three groups: those least concerned about AI extinction risk ($n = 53$, whose forecasts were $\leq 0.25\%$), those in the middle ($n = 54$, whose forecasts were $\geq 0.25\%$ and $\leq 4.5\%$), and those most concerned about AI extinction risk ($n = 53$, whose forecasts were $\geq 4.5\%$). For each group, we computed the average absolute distance from the (imputed) tournament median in standard deviations on questions resolving in 2024, 2030, 2050, and

2100. Fig. 4 shows the medians at each resolution date, accompanied by bootstrapped confidence intervals.

From a motivated cognition perspective, forecasters with strong pre-tournament beliefs on existential risk should make more extreme forecasts on long-run questions (resolving in 2050/2100) than on questions resolving in 2024 and 2030. We see the strongest evidence for a pattern of motivated cognition among forecasters most concerned about AI extinction risk. Relative to the least concerned and moderate forecasters, those most concerned about AI extinction risk staked out more extreme positions on later resolution dates than on closer dates. The least concerned and moderates did not differ from each other to the same degree, providing forecasts that are, if anything, more in line with group medians at longer time horizons (2100) relative to shorter time horizons (2024). Fig. 4 shows that the least concerned and moderately concerned forecasters decreased in distance

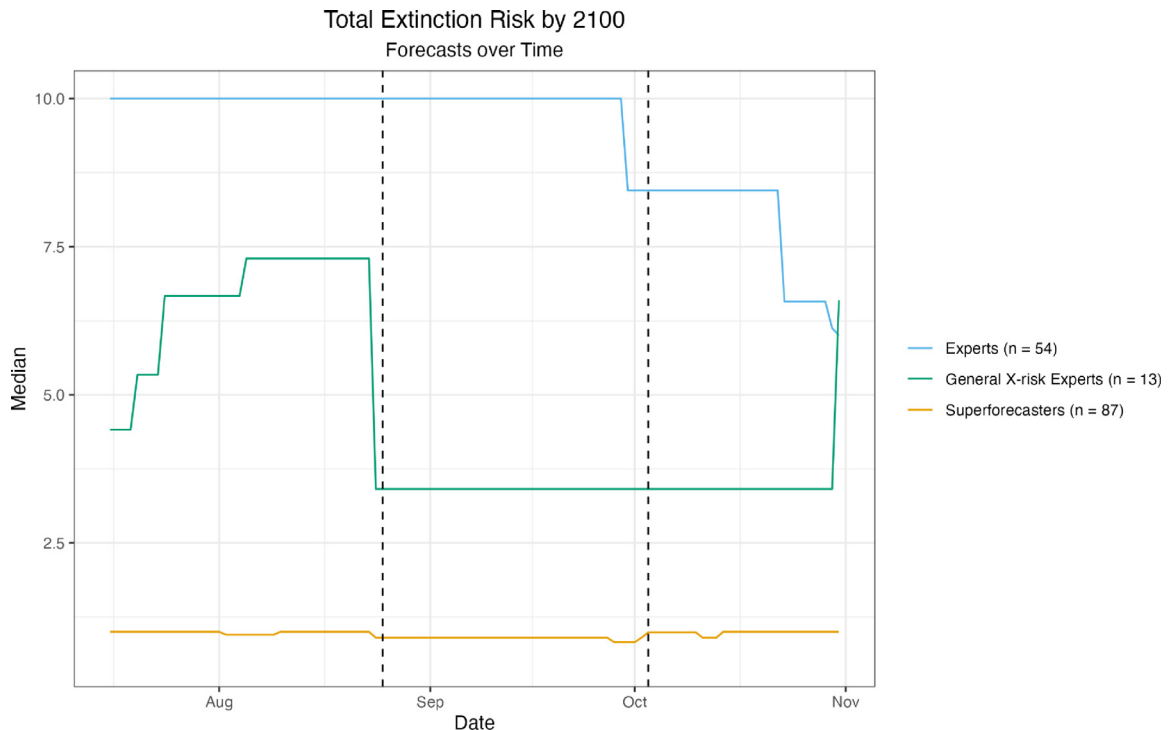


Fig. 2. The median forecast for total extinction risk (by 2100) during all group stages of the XPT for three groups who took part in the XPT: superforecasters, experts, and risk experts (who specialize in topics related to existential risks, a subset of all experts). Dotted lines in the above graph denote the separation of Stage 2 (on the left), Stage 3 (in the middle) and Stage 4 (on the right). During Stage 2, superforecasters and experts had access to only in-group forecasts and rationales; during Stages 3 and 4, groups could see each others' forecasts and rationales. The figure shows that the superforecaster median stayed relatively stable during the tournament, and that while different groups of experts converged toward one another during Stage 2, they did not converge toward the group of superforecasters. Many forecasters chose to update their numeric forecasts toward the end of this stage, hence the sharp reduction of the expert and risk expert medians around this time.

from the median for further-out forecasts whereas the most concerned forecasters exhibited the opposite tendency. See [Appendix](#) and [Fig. A.4](#) for additional analyses of this data.

The motivated cognition pattern was less pronounced for extinction-risk forecasts in non-AI risk areas ([Karger et al., 2023](#)). For example, when we split forecasters by their degree of concern about nuclear extinction ([Fig. 5](#)), the most concerned diverge more from the median on all questions than the less concerned but are not more extreme on longer vs. shorter time-horizon questions. For biorisk, evidence for the extremization of long-run forecasts only emerged in muted form.

To explore the connections between extreme risk forecasts and skill at predicting the forecasts of others (perspective-taking) we used the Reciprocal Scoring measure (see [Methods](#) section; [Karger et al. \(2021\)](#)), which asks each forecaster to (independently) predict the predictions of other forecasters. Participants gave their own forecast and two reciprocal predictions for each required question about existential and catastrophic risk: one reciprocal prediction asked what the median superforecaster would say, and the other asked what the median

expert would say. Forecasters did these tasks before seeing other participants' forecasts. We computed intersubjective accuracy by separately comparing each forecaster's prediction of others' forecasts on 10 key questions, for both experts and superforecasters. We ranked each forecaster's accuracy on those 20 quantities relative to other participants, and then we computed each forecaster's average rank to calculate an overall measure of accuracy.

[Fig. 6](#) plots total extinction risk by 2100 for each group and displays quintiles of reciprocal scoring accuracy (most accurate on the left, lowest on the right). Here, we find that participants who were the best at perspective-taking (as measured by accuracy in forecasting other groups' predictions) were less concerned about existential risk than those who were worse at perspective-taking. This was true among both specialists and generalists/superforecasters. Put differently, the better experts were at discerning what other people would predict, the less concerned they were about extinction. In [Fig. A.5](#), we see the same pattern for catastrophic risk: superforecasters and experts who performed better on reciprocal scoring tasks estimate a lower chance of catastrophe.

For additional commentary and figures related to reciprocal scoring accuracy, see [Appendix, Figs. A.5–A.8](#).

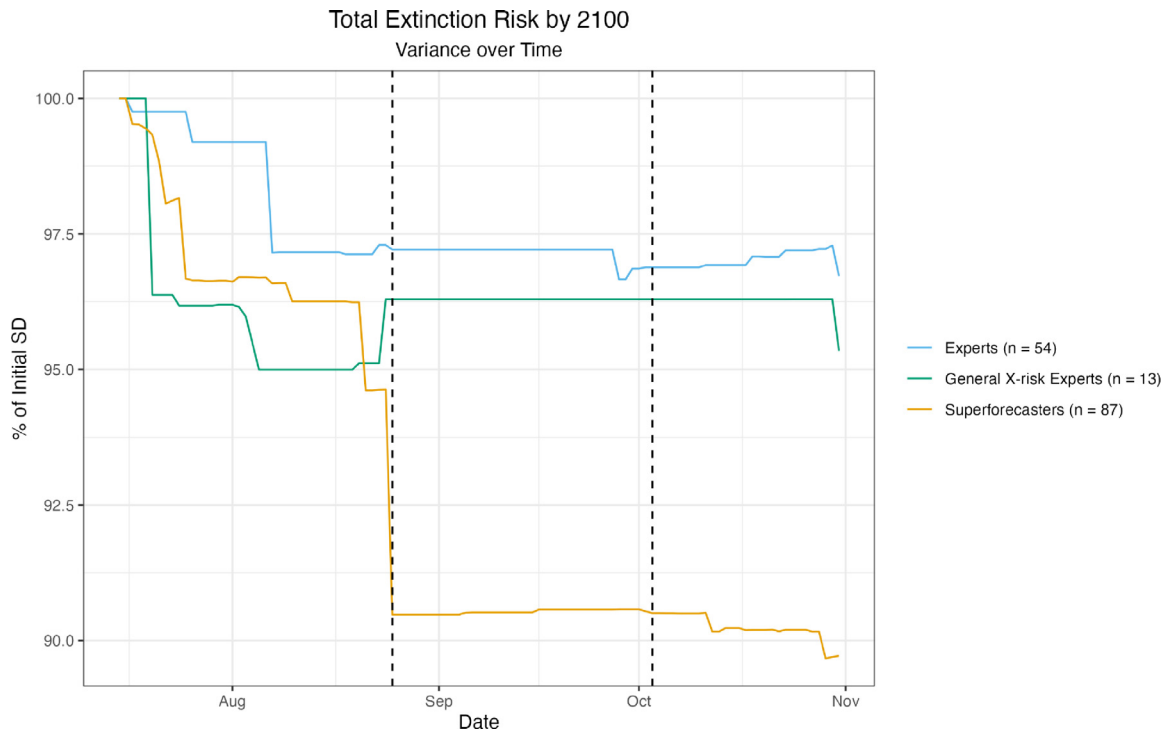


Fig. 3. The intra-group variance of forecasts for total extinction risk (by 2100) during the three public-deliberation stages of the XPT for three groups: generalists/superforecasters, subject-matter experts working on specialty topics, and experts working outside their field. The amount of variance within groups, as pegged to the variance of initial forecasts in Stage 1, dropped only slightly across the tournament, with superforecasters converging toward one another slightly more than expert groups. Many forecasters chose to update their numeric forecasts toward the end of this stage, hence the sharp changes in variance around this time.

6. Discussion

We focus our discussion on three main areas: (a) content issues: what forecasters thought about existential risks; (b) process issues: what lessons we can draw about the difficulty of changing views across the four stages of the XPT; (c) intersubjective metrics: their potential to serve as lead indicators of which forecasters are most likely to be accurate.

On **content**, we found a consistent and persistent divide between subject-matter specialists and talented generalists in which: (a) experts put higher probabilities on both catastrophic and existential threats; (b) this pattern was more pronounced for existential threats, especially those with anthropogenic and AI causation. These data raise at least three sets of distinct but overlapping questions.

First, why do the subject-matter specialists quite consistently rate both catastrophic and existential risks higher than the generalists/superforecasters do? The error-and-bias school of thought—especially [Kahneman's 2011](#) work on focalism—implies one answer. The more time we spend thinking about something, the more important it becomes in our minds. And, of course, experts have devoted far more time than the generalists to their

specialties. An alternative rationalist interpretation simply posits that the experts are far better informed. These two perspectives should be testable over the next several years as forecasting accuracy data for the short-run XPT questions becomes available. The Kahneman perspective implies that, when accuracy data arrive, we should expect either zero or negative correlations between subject-matter expertise and accuracy across fields, whereas the rationalist view implies that we should expect positive correlations.

Second, why were the specialist-generalist disagreements much more muted when judging non-anthropogenic causes of catastrophic or existential risk? Neither of the two explanations invoked above—the Kahneman focalism argument nor the rationalist argument—is well prepared to explain this exception. One possibility open to future research is that anthropogenic debates, relative to non-anthropogenic debates, are more emotional and easily politicized because they imply attributions of blame to specific human actors: corporations, governments, or individual actors.

Third, although it is not surprising that forecasters across the board judged the less extreme catastrophic risks (scenarios killing only 10% of the population) as more likely than existential risks (wiping out humanity),

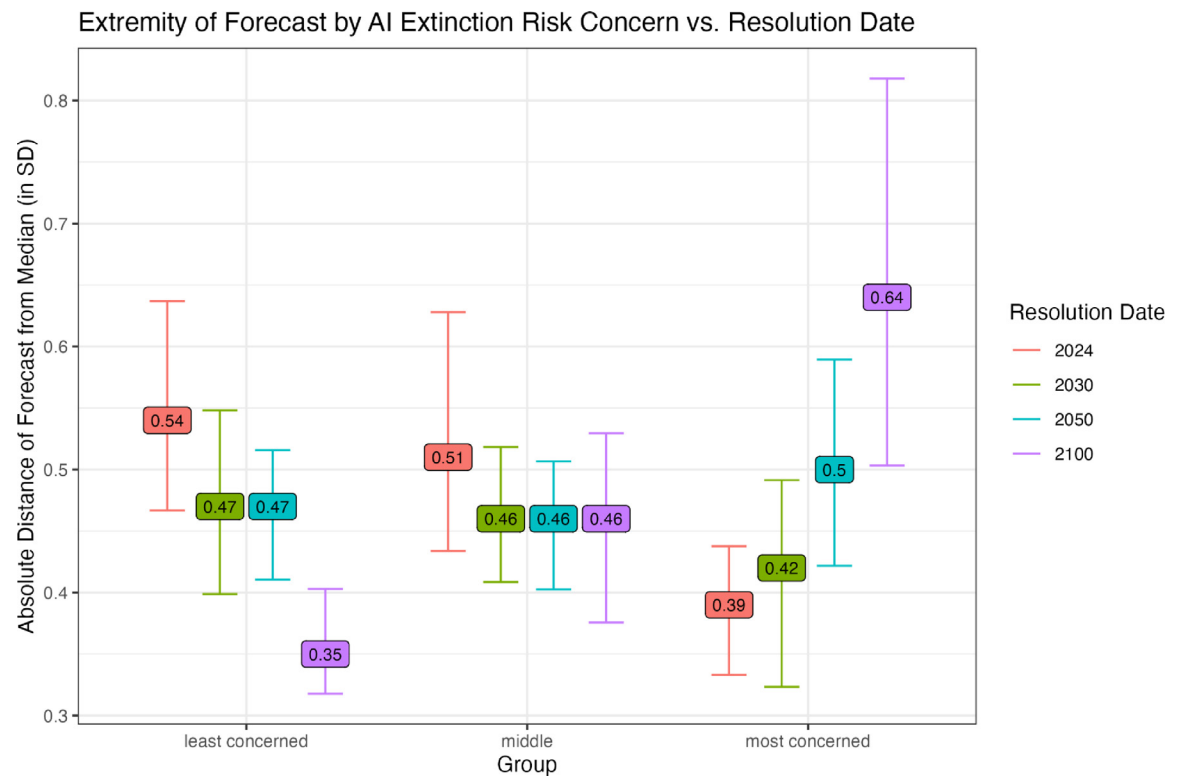


Fig. 4. The extremity of shorter- vs. longer-range AI-risk forecasts of forecasters who had pre-tournament views of varying extremity.

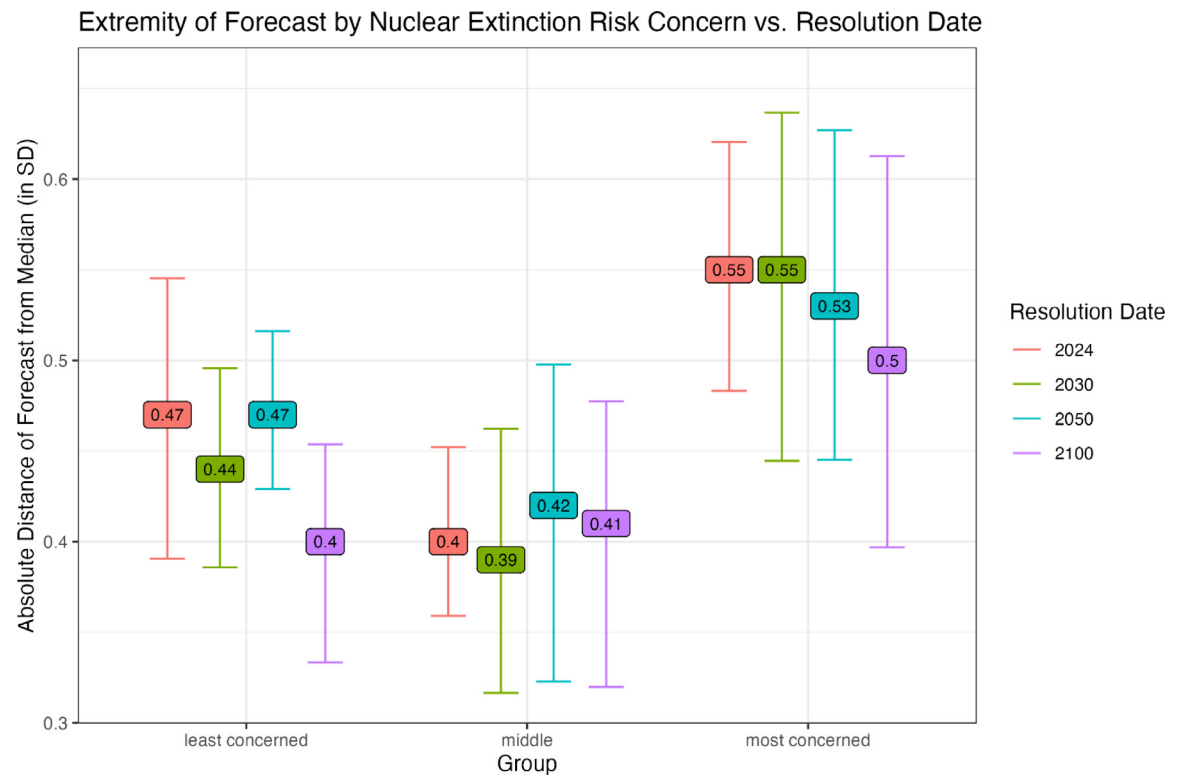


Fig. 5. The extremity of short- vs. long-range nuclear-extinction forecasts of forecasters with pre-tournament views of varying extremity.

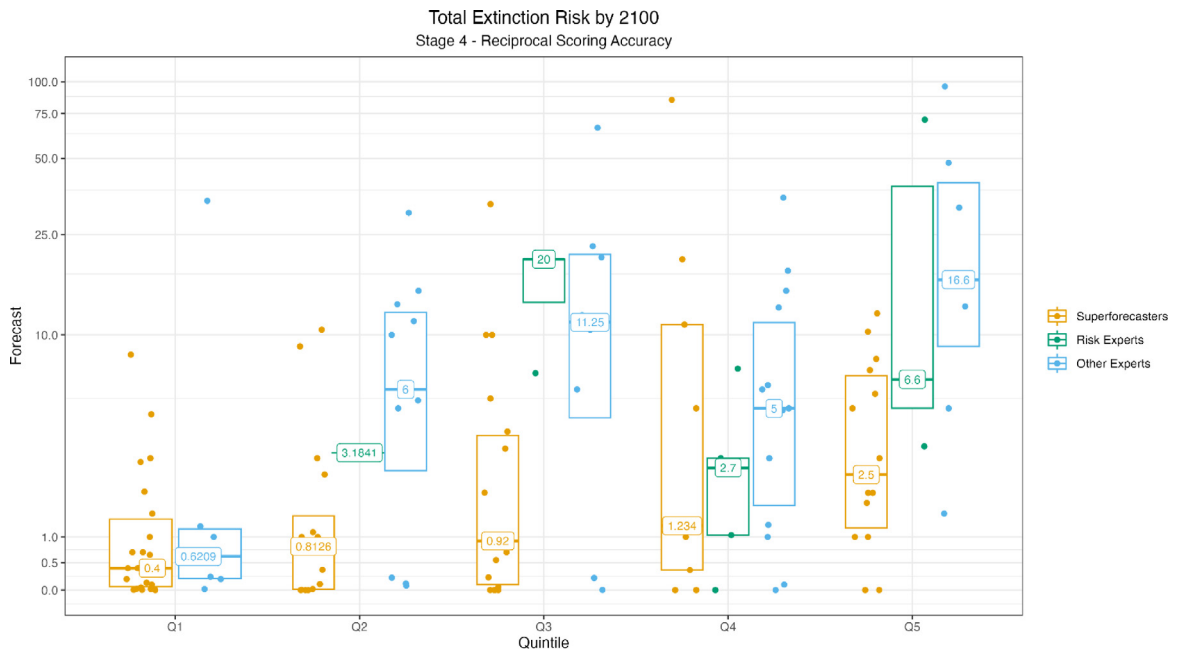


Fig. 6. The relationship between final forecasts and reciprocal scoring accuracy for superforecasters, risk experts, and other experts. The x-axis is reciprocal accuracy quintiles (Q1 is the top 20% of forecasters; Q5, the bottom 20%). The y-axis plots the final Stage 4 forecasts on the XPT question of whether humanity will go extinct by 2100. Note that the y-axis is non-standard and uses a pseudo-log transformation to stretch the 1–10% range to visualize the key disagreements. No Risk Experts are shown in Q1 because there were none in that quintile.

why did the divide between specialist and generalist opinion grow widest when it came to judgments of long-run AI-induced existential risk? One possibility is that humans have had direct experience with nuclear weapons, pandemics, and climate change but no interactions, as of 2024, with super-intelligent AIs. Perhaps only AI experts have the requisite knowledge to fill in the unknowns and see the merits in AI-risk arguments that generalists often find far-fetched. This account implies the specialist-generalist gap will narrow as we get closer to achieving super-intelligent machines, which seems to go against the general thesis that expertise increases bias in forecasting.

On **process**, what research implications should we draw from the lack of opinion movement across the four stages and the four months of rather intense information exchange among participants?

One view counsels patience. It attributes the lack of opinion convergence to lack of diagnostic signals, especially for possible futures extending to the end of the century. The best arguments made by participants in the tournament were just not good enough. Instead, forecasters had to rely on internalized priors in the absence of external evidence, and this was particularly unavoidable on the most speculative topic: AI. In this view, if opinion convergence ever occurs, it will be only after early warning indicator questions resolve and induce influential members of each camp to update their forecasts in

proportion to the perceived relevance of the early indicators. This view could be bolstered by further analysis of forecasters' rationales and arguments from the XPT.

Another view is less patient. It attributes the lack of opinion convergence to methodological shortcomings of the XPT, particularly inadequate incentives for forecasters to engage in adversarial collaborations (Kahneman, 2011) and an incentive structure that makes ducking tough questions from the other side or talking past each other much easier (Tetlock, Karvetski, Satopää, & Chen, 2023). It is also possible that other methodological shortcomings prevented opinion convergence such as time available to submit forecasts, differing levels of activity across forecasting teams, etc.; or that other psychological phenomena like egocentric discounting, often observed in judge-advisor systems, strongly discourage forecasters from changing their minds even when opposing views are credible (Bonaccio & Dalal, 2006; Yaniv & Kleinberger, 2000).

One question is whether the XPT would have yielded more evidence of opinion convergence if it had been designed more like a Delphi process, with anonymity (e.g., randomly generated usernames, norms against participants naming themselves in discussions) and more tightly controlled feedback (for a review of Delphi approaches, see Rowe and Wright (1999)). The XPT already bears some family resemblance to Delphi, given the independent forecasts in Stage 1 that participants brought

to the group discussions. The Delphi family of methods is designed to encourage participants to focus on the substantive merits of arguments for and against points of view, regardless of the status of the person advancing the arguments. Both the typical Delphi method and the XPT attempt to do this by initially eliciting judgments and rationales under a cloak of anonymity before sharing them with others. However, the XPT deviated from the Delphi process because it disclosed participants' judgments as soon as forecasters entered Stage 2, whereas, in Delphi, organizers preserve anonymity into the second stage—and indeed may preserve anonymity much longer through a number of iterations. We did not adopt an extended Delphi process for the XPT because we wanted to jumpstart debates and adversarial collaborations among forecasters as soon as practical.

But that judgment call may have been a mistake. People find it easier to change their minds when they are not anchored down by past public commitments to positions (Festinger, 1957), and they also find it easier to change their minds if they can do so in privacy, reducing the risk of appearing weak or confused. The next operationalization of the XPT might do well to encourage or require the expression of anonymous judgments throughout all four stages of deliberation.

Although the data at hand in 2023 cannot tell us whose judgments about 2030, 2050, or 2100 to trust, the XPT does offer clues in its intersubjective measure of forecasting accuracy: Reciprocal Scoring. Within each group—specialists and generalists—a strong negative relationship emerged between accuracy at predicting others' forecasts and the extremity of existential risk probability estimates, especially long-run AI risk. Extreme AI-risk forecasters made particularly extreme long-range forecasts for both objective outcomes (what will actually happen in 2100?) and intersubjective outcomes (what do others think will happen in 2100?). This pattern suggests that the pre-tournament extreme forecasters actually believed in their extreme long-range forecasts and were not just posturing to call attention to a cause they want dangerously complacent generalists/superforecasters to take seriously. The AI-risk extreme forecasters did not tactically retreat from extremism when the prospect of embarrassment was imminent because their intersubjective forecasts were falsifiable in the short term. One plausible explanation is “naïve realism”: forecasters who saw large long-run AI risks assume intelligent observers see the same world and draw similar conclusions—the same egocentric judgment process conjectured to underlie the false consensus effect (Gilovich, 1990; Ross et al., 1977).

That said, a cautionary note is in order. Although it is suggestive that the correlates of extremism and resistance to viewpoint convergence in the XPT are also correlates of worse forecasting performance,¹⁶ earlier work did not test intersubjective metrics like Reciprocal Scoring on long-range existential questions. So, whether holders of extreme views in the 2023 run of the XPT will also prove

less accurate in 2030, 2050, or 2100 remains an open empirical question. To test changes in belief and persuadability, we plan to track the beliefs of these participants over the next decade using follow-up surveys to measure belief-updating as we begin to provide accuracy feedback to participants.

Putting aside the currently unresolvable question of who is right, the data do highlight the current intractability of the debates. Across topics, we see the inability of each side to sway the other's opinions, despite strong incentives to share their best arguments in a four-month, multi-stage tournament. In a detailed analysis of verbatim argumentation, Karger et al. (2023) have shown how easy it is even for well-educated, truth-seeking parties to talk past each other. On AI, for instance, the least and most concerned camps could not even agree on who bore the burden of proof. Although both endorsed the old maxim “extraordinary claims require extraordinary evidence”, they disagreed over who was making the extraordinary claims. The least concerned camp saw no base-rate precedents for worst-case AI scenarios, so they saw probabilities of 50/50 or higher for AI-induced extinction events by 2100 as far-fetched. The most concerned camp was baffled by why the other camp was ignoring the most relevant base-rate precedent: what happens when technologically superior societies collide with less advanced societies? The most concerned camp saw an exponential take-off trajectory for AIs (perhaps AIs designing super-dangerous viruses) that could overwhelm humanity astonishingly fast and saw the other camp as unimaginative and too grounded in incremental status quo projections.

All of this suggests that absent dramatic early warning signals, debates over existential risks are difficult to resolve, particularly in contexts where participants have strong priors. Future research to address the intractability of these debates might focus on restructuring the forecasting process to encourage ongoing engagement from representatives of different schools of thought, more tightly controlling discussion norms and maintaining forecaster anonymity, or finding better ways to incentivize honest perspective-taking.

CRediT authorship contribution statement

Ezra Karger: Writing – review & editing, Writing – original draft, Validation, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Josh Rosenberg:** Writing – review & editing, Methodology, Investigation, Conceptualization. **Zachary Jacobs:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Investigation, Formal analysis, Data curation. **Molly Hickman:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Formal analysis, Data curation. **Phillip E. Tetlock:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Funding acquisition, Conceptualization.

¹⁶ Pavel Atanasov et al. “Full Accuracy Scoring Accelerates the Discovery of Skilled Forecasters”, *SSRN Working Paper* (February 14, 2023), <http://dx.doi.org/10.2139/ssrn.4357367>.

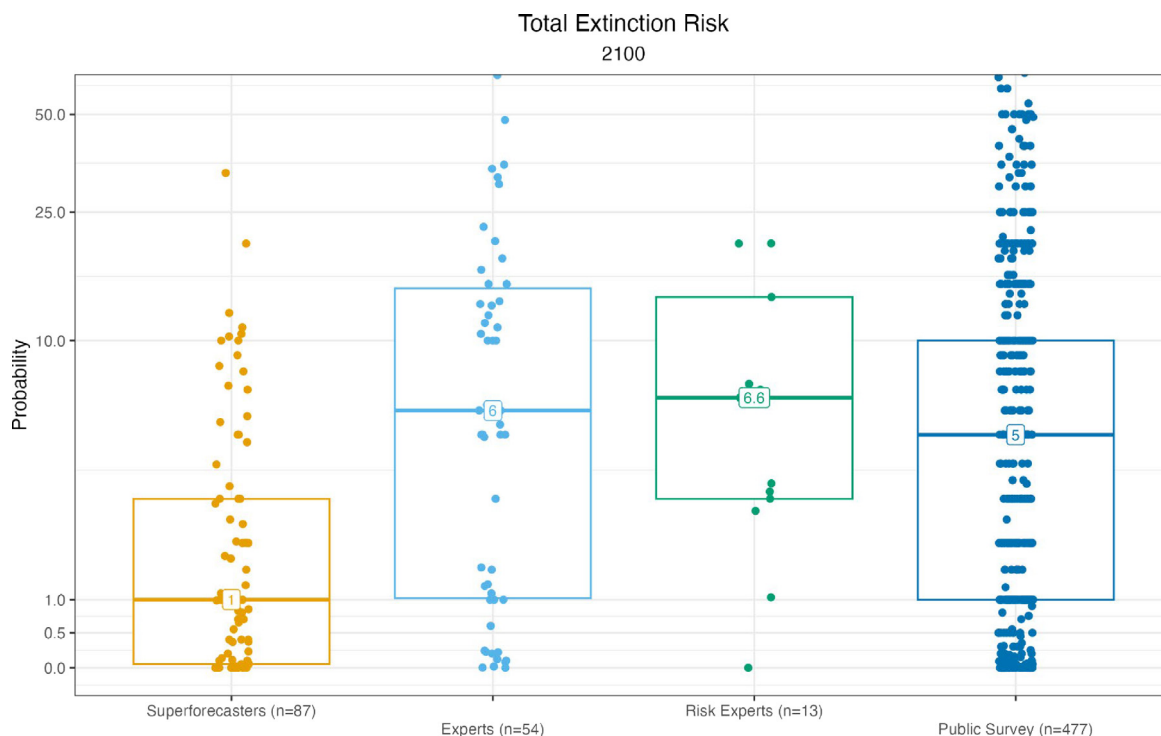


Fig. A.1. The distribution of final Stage 4 forecasts of extinction risk (by 2100) for each of three groups in the XPT (superforecasters, domain experts, and risk experts) as well as a public survey of college-educated respondents from an online platform. Box plots show the 25–75th percentile range (boxes) and the median (labeled) for each group. Within each group, points are jittered slightly horizontally to show density, but this jittering has no other empirical purpose.¹⁷

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix. Additional figures

This section contains figures the authors believe provide valuable context, but do not necessarily share the same level of priority as those presented in the main text.

To visualize the large variation in beliefs among subgroups, Fig. A.1 shows each forecast of total extinction risk, from XPT forecasters and from our parallel public survey, with box plots indicating the 25th and 75th percentile ranges in each group. Here, even among the groups that assign the lowest risk (superforecasters) and highest risk (general x-risk experts), we can see significant overlap in the distribution of forecasts.

Figs. A.2 and A.3 show the same pattern for forecasts of total catastrophic risk and of AI extinction risk.

The result shown in Fig. 4 is not an artifact of trichotomous binning: Fig. A.4 presents the same data continuously, with the x-axis indicating forecasts on AI extinction risk by participant and the y-axis measuring differences in standard deviation units from the median, with questions split into four resolution-year categories.

Fig. A.5 shows the same pattern for extinction risk from AI specifically as Figure 10 did.

Fig. A.6 uses a continuous scale to directly plot the forecasts of participants on AI risk and their Reciprocal Scoring accuracy. This correlation is significant ($r = 0.22$, $p = .007$).

Fig. A.7 displays the direction of the errors driving the lower Reciprocal Scoring accuracy scores of forecasters most concerned about AI risk. An asymmetric false-consensus effect emerged. Forecasters least concerned about AI risk (those who made estimates of extinction risk by 2100 below 1%) slightly underestimated the level of

¹⁷ The y-axes on box plots have been truncated above at the 95th percentile of participants + a 5% cushion. This means that we exclude some high outlying forecasts from at most (but usually many less

than) 5% of tournament participants. This also excludes a portion of public survey respondents (those whose forecasts are above the 95th percentile of XPT participant forecasts).

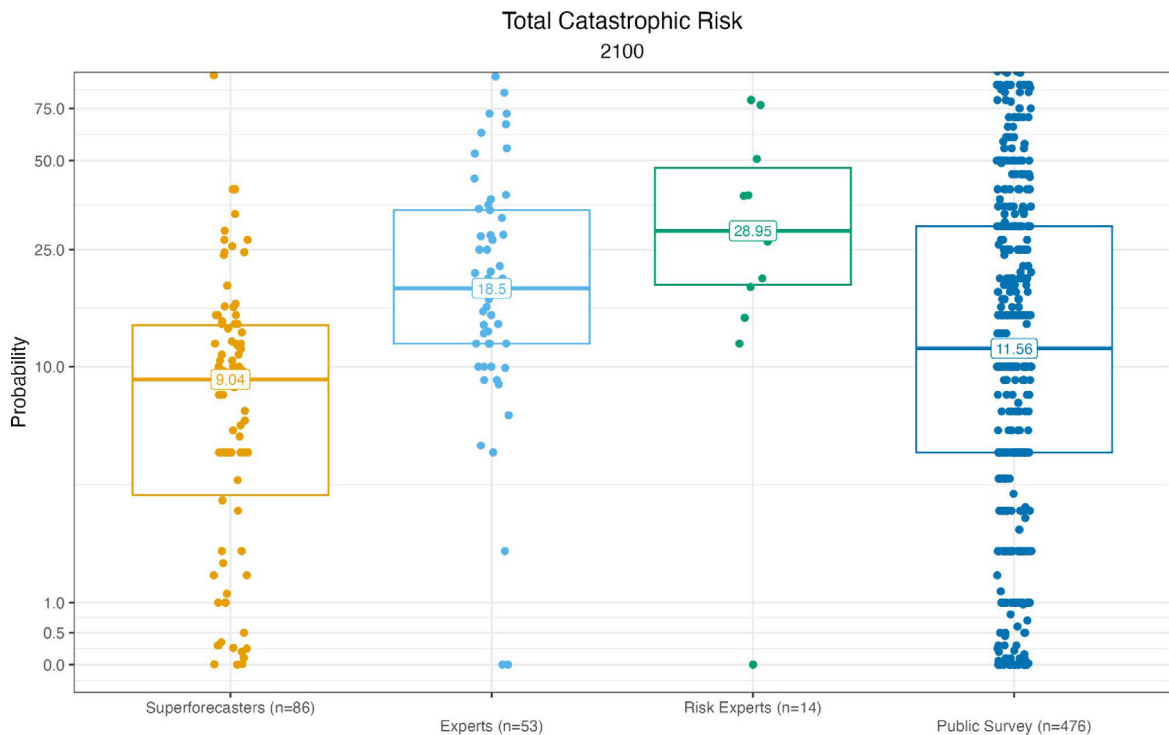


Fig. A.2. The distribution of final Stage 4 forecasts of catastrophic risk (by 2100) for superforecasters, domain experts, and risk experts as well as a public survey of college-educated respondents. Box plots show the 25–75th percentile range (boxes) and the median (labeled) for each group. Within each group, points are jittered slightly horizontally to show density.

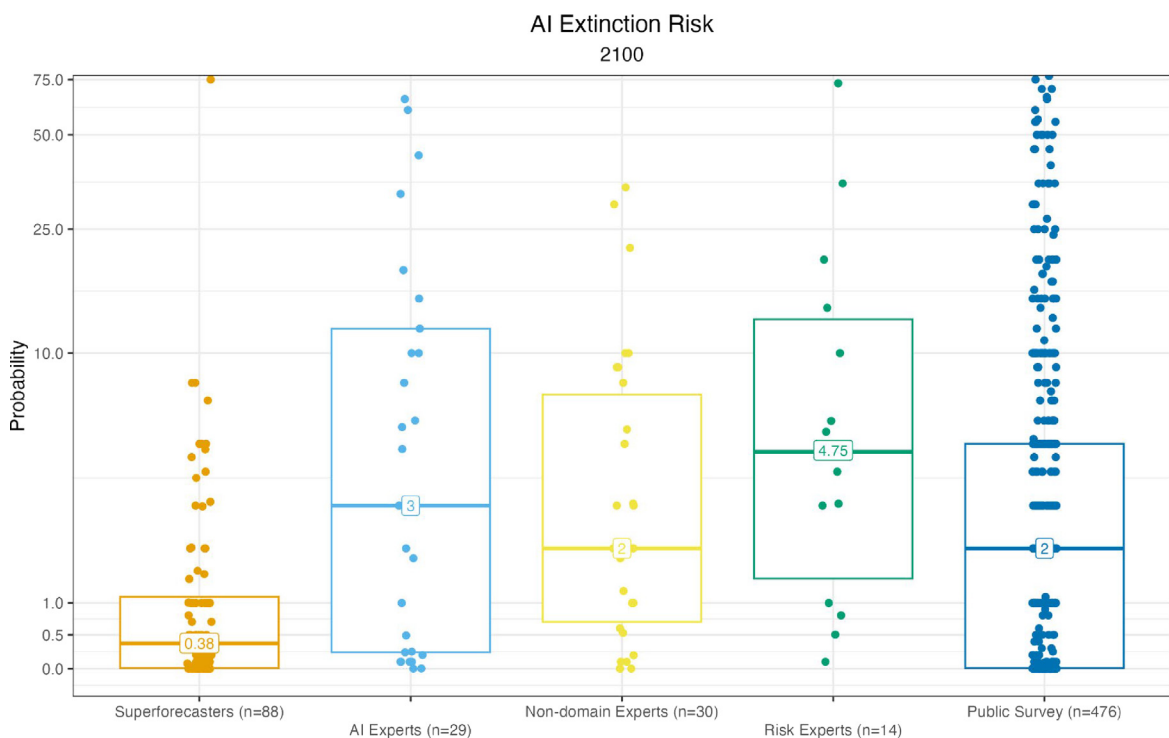


Fig. A.3. The distribution of final forecasts of AI extinction risk (by 2100) for each of four groups in the XPT (superforecasters, AI experts, non-AI experts, and risk experts) as well as a public survey of college-educated respondents. Box plots show the 25–75th percentile range (boxes) and the median (labeled) for each group. Within each group, points are jittered slightly horizontally to show density.

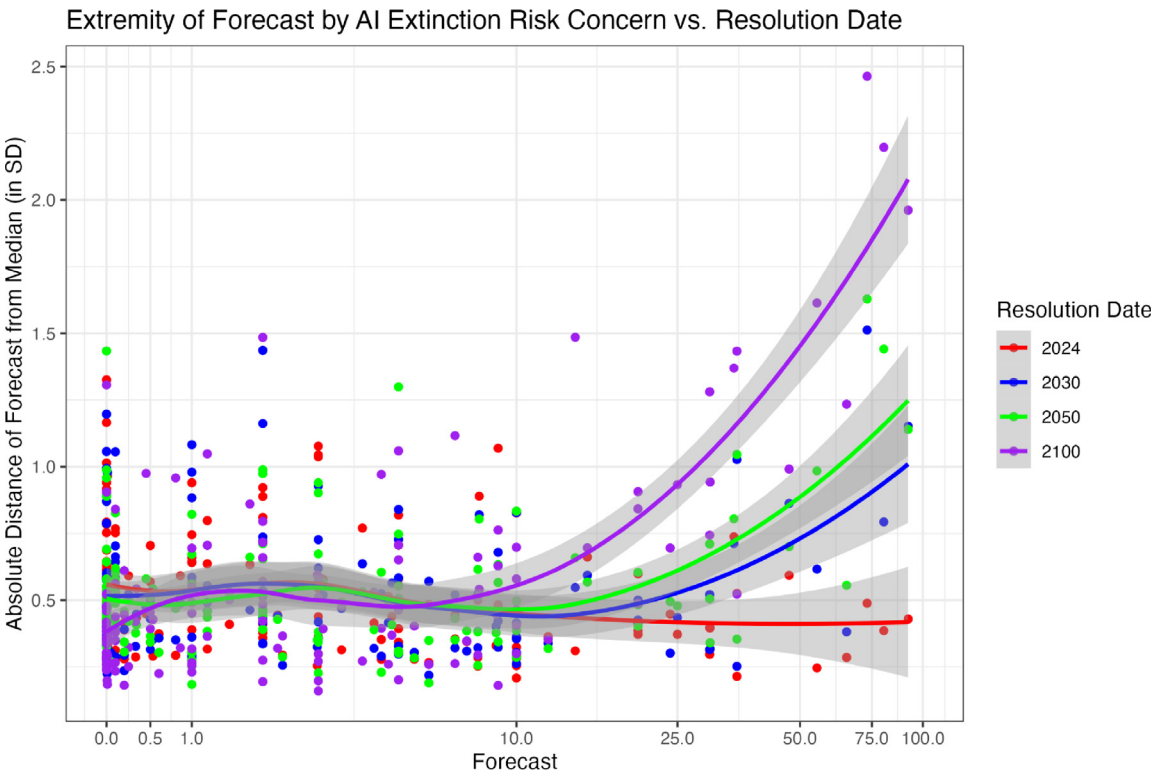


Fig. A.4. Each participant's forecast of AI extinction risk by 2100 on the x-axis and, on the y-axis, shows their mean absolute difference from the median on questions resolving in 2024, 2030, 2050, and 2100 in standard deviation units. Note that each forecaster is represented by four points on the graph, representing each of the resolution years.

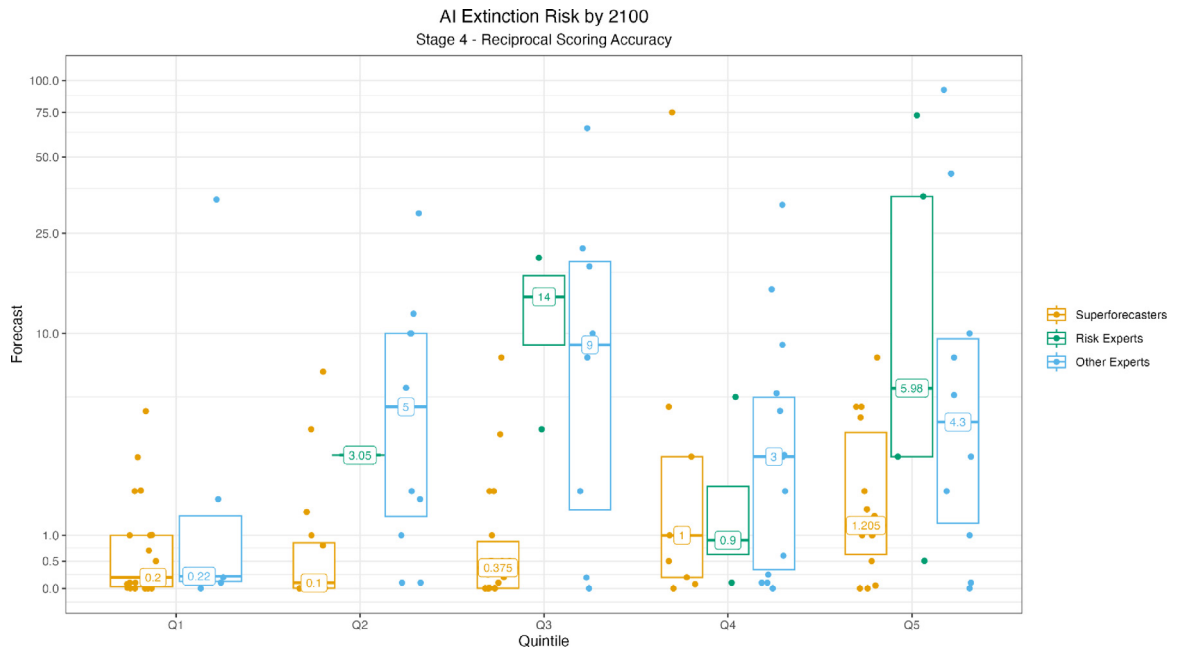


Fig. A.5. The relationship between final forecasts and reciprocal scoring accuracy for each group (superforecasters, risk experts, and other experts). The x-axis is again reciprocal accuracy quintiles. The y-axis plots the final (Stage 4) forecasts on extinction risk due to AI by 2100.

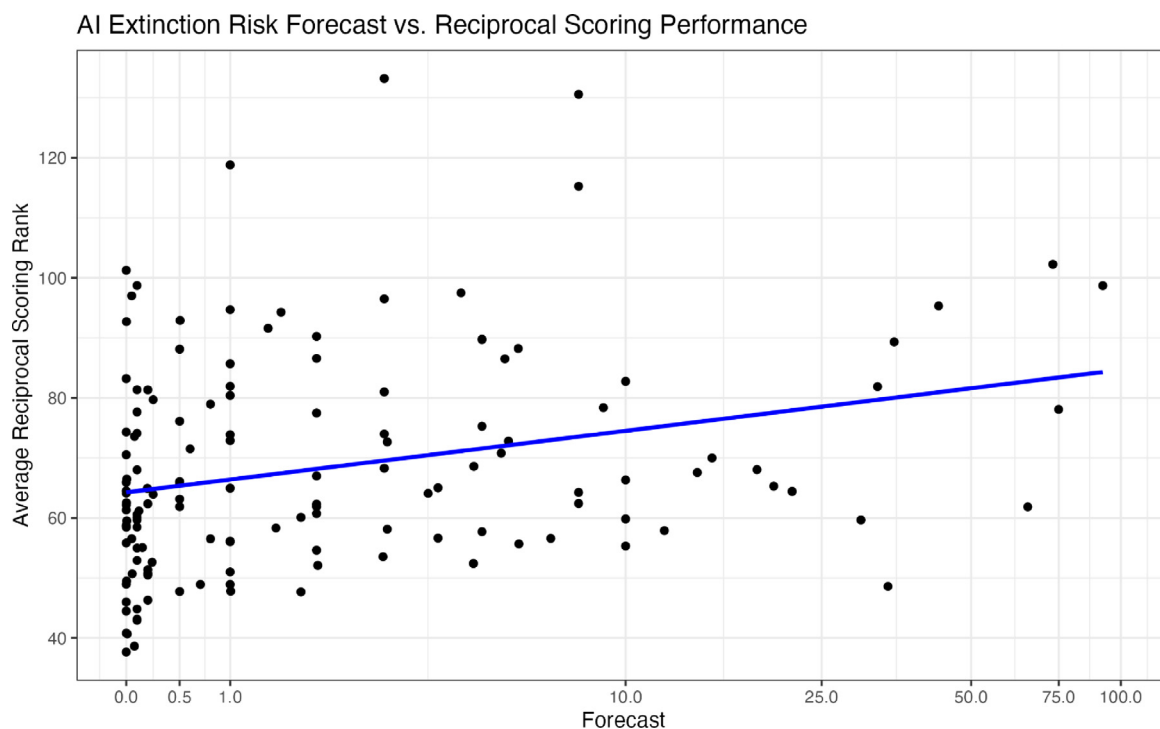


Fig. A.6. The relationship between each participant's forecast of AI extinction risk by 2100, on the x-axis, and their reciprocal scoring performance. For each risk-related question, participants were asked to provide their own true beliefs as well as a series of intersubjective forecasts of generalist/superforecasters and domain specialists' views, and were ranked based on their proximity to the true median forecasts of these groups. The y-axis values capture the mean rank of each forecaster on these questions.

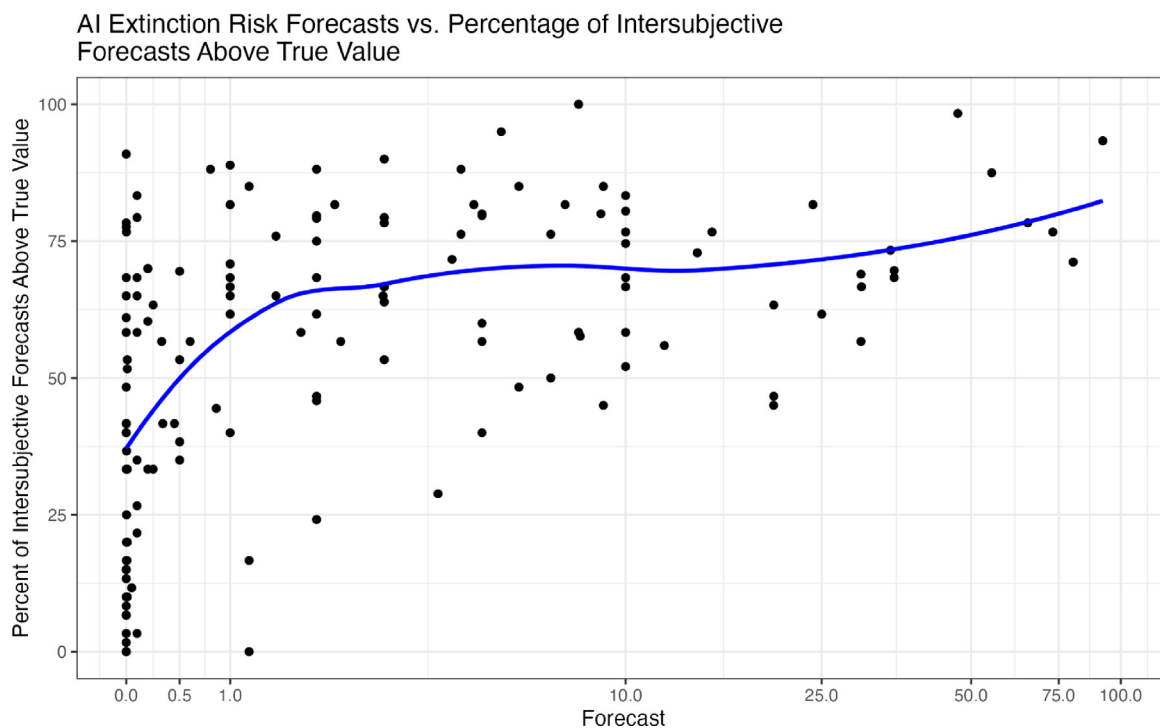


Fig. A.7. The relationship between each forecasters' own forecast of AI Extinction Risk by 2100 (on the x-axis) and the percentage of their intersubjective forecasts (i.e. their forecasts of superforecasters and domain experts) that were above the true median forecasts provided by those groups.

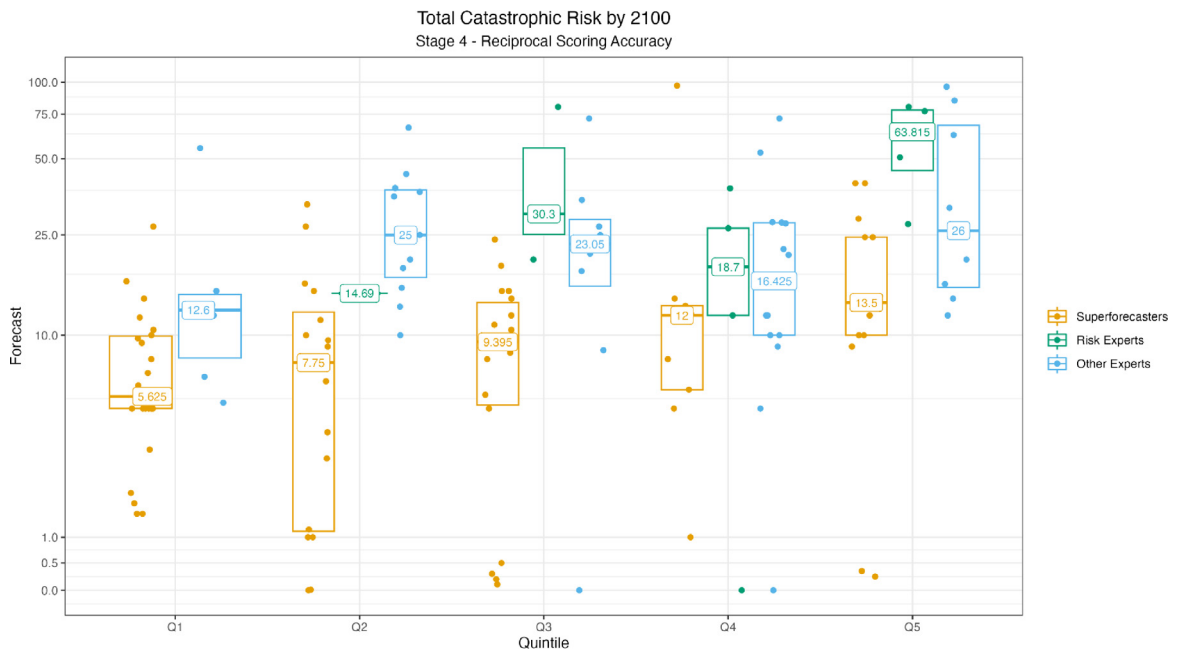


Fig. A.8. The relationship between final forecasts and reciprocal scoring accuracy (i.e., how accurately did the forecaster predict the median of the superforecaster judgments and of the expert judgments) for each group (superforecasters, risk experts, and other experts). The x-axis is reciprocal accuracy quintiles (Q1 is the top 20% of forecasters; Q5, the bottom 20%)(Q1 is the fifth of forecasters whose reciprocal scoring accuracy was best; Q5 is the fifth of forecasters whose reciprocal scoring accuracy was worst). The y-axis plots the final (Stage 4) forecasts on catastrophic risk by 2100.

concern among other forecasters, but this tendency was 34% of the magnitude of the over-estimation observed among forecasters who made probability estimates above 1%. Similar patterns emerge for each other category of existential risk. See Fig. A.8.

Data availability

Code and data to replicate tables, figures and data analysis are provided as part of a replication package. The replication package is available at <https://github.com/forecastingresearch/xpt-ijf-replication>. This package is adapted from a previously-published repository associated with initial reporting on the XPT, available at <https://github.com/forecastingresearch/xpt-lib>.

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